

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("cardio_train_data.csv", sep=';')
```

```
In [6]: print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
```

```
   id  age  gender  height  weight  ap_hi  ap_lo  cholesterol  gluc  smoke  \
0   0  18393      2    168   62.0   110    80           1      1      0
1   1  20228      1    156   85.0   140    90           3      1      0
2   2  18857      1    165   64.0   130    70           3      1      0
3   3  17623      2    169   82.0   150   100           1      1      0
4   4  17474      1    156   56.0   100    60           1      1      0
```

```
   alco  active  cardio
0      0        1        0
1      0        1        1
2      0        0        1
3      0        1        1
4      0        0        0
```

```
(70000, 13)
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 70000 entries, 0 to 69999
```

```
Data columns (total 13 columns):
```

#	Column	Non-Null Count	Dtype
0	id	70000 non-null	int64
1	age	70000 non-null	int64
2	gender	70000 non-null	int64
3	height	70000 non-null	int64
4	weight	70000 non-null	float64
5	ap_hi	70000 non-null	int64
6	ap_lo	70000 non-null	int64
7	cholesterol	70000 non-null	int64
8	gluc	70000 non-null	int64
9	smoke	70000 non-null	int64
10	alco	70000 non-null	int64
11	active	70000 non-null	int64
12	cardio	70000 non-null	int64

```
dtypes: float64(1), int64(12)
```

```
memory usage: 6.9 MB
```

```
None
```

	id	age	gender	height	weight
count	70000.000000	70000.000000	70000.000000	70000.000000	70000.000000
mean	49972.419900	19468.865814	1.349571	164.359229	74.205690
std	28851.302323	2467.251667	0.476838	8.210126	14.395757
min	0.000000	10798.000000	1.000000	55.000000	10.000000
25%	25006.750000	17664.000000	1.000000	159.000000	65.000000
50%	50001.500000	19703.000000	1.000000	165.000000	72.000000
75%	74889.250000	21327.000000	2.000000	170.000000	82.000000
max	99999.000000	23713.000000	2.000000	250.000000	200.000000

	ap_hi	ap_lo	cholesterol	gluc	smoke
count	70000.000000	70000.000000	70000.000000	70000.000000	70000.000000
mean	128.817286	96.630414	1.366871	1.226457	0.088129
std	154.011419	188.472530	0.680250	0.572270	0.283484
min	-150.000000	-70.000000	1.000000	1.000000	0.000000
25%	120.000000	80.000000	1.000000	1.000000	0.000000
50%	120.000000	80.000000	1.000000	1.000000	0.000000
75%	140.000000	90.000000	2.000000	1.000000	0.000000
max	16020.000000	11000.000000	3.000000	3.000000	1.000000

	alco	active	cardio
count	70000.000000	70000.000000	70000.000000
mean	0.053771	0.803729	0.499700
std	0.225568	0.397179	0.500003
min	0.000000	0.000000	0.000000
25%	0.000000	1.000000	0.000000
50%	0.000000	1.000000	0.000000
75%	0.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000

```
In [7]: print(df.isnull().sum())
```

```
id          0
age         0
gender      0
height      0
weight      0
ap_hi       0
ap_lo       0
cholesterol 0
gluc        0
smoke       0
alco        0
active      0
cardio      0
dtype: int64
```

```
In [8]: df = df.drop_duplicates()
```

```
In [9]: df['age_years'] = df['age'] / 365
```

```
In [10]: df['BMI'] = df['weight'] / ((df['height'] / 100) ** 2)
```

```
In [11]: print("Average BMI:", df['BMI'].mean())
```

Average BMI: 27.55651255127886

```
In [12]: high_bp = df[(df['ap_hi'] >= 140) | (df['ap_lo'] >= 90)]
percentage_bp = (len(high_bp) / len(df)) * 100

print("High BP Percentage:", percentage_bp)
```

High BP Percentage: 35.292857142857144

```
In [13]: high_chol = df[df['cholesterol'] > 1]
percentage_chol = (len(high_chol) / len(df)) * 100

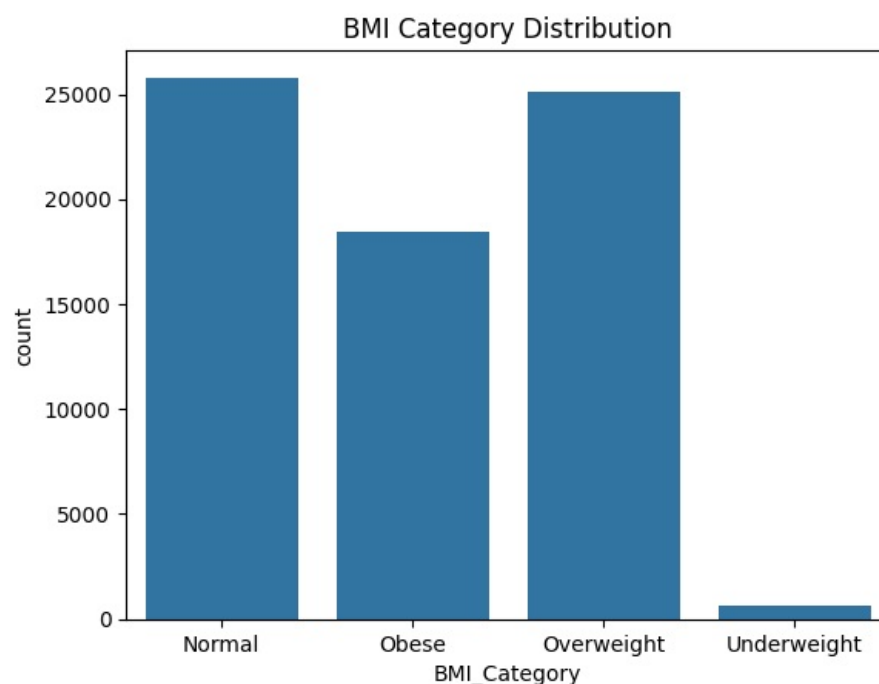
print("High Cholesterol %:", percentage_chol)
```

High Cholesterol %: 25.164285714285718

```
In [14]: def bmi_category(bmi):
    if bmi < 18.5:
        return "Underweight"
    elif bmi < 25:
        return "Normal"
    elif bmi < 30:
        return "Overweight"
    else:
        return "Obese"

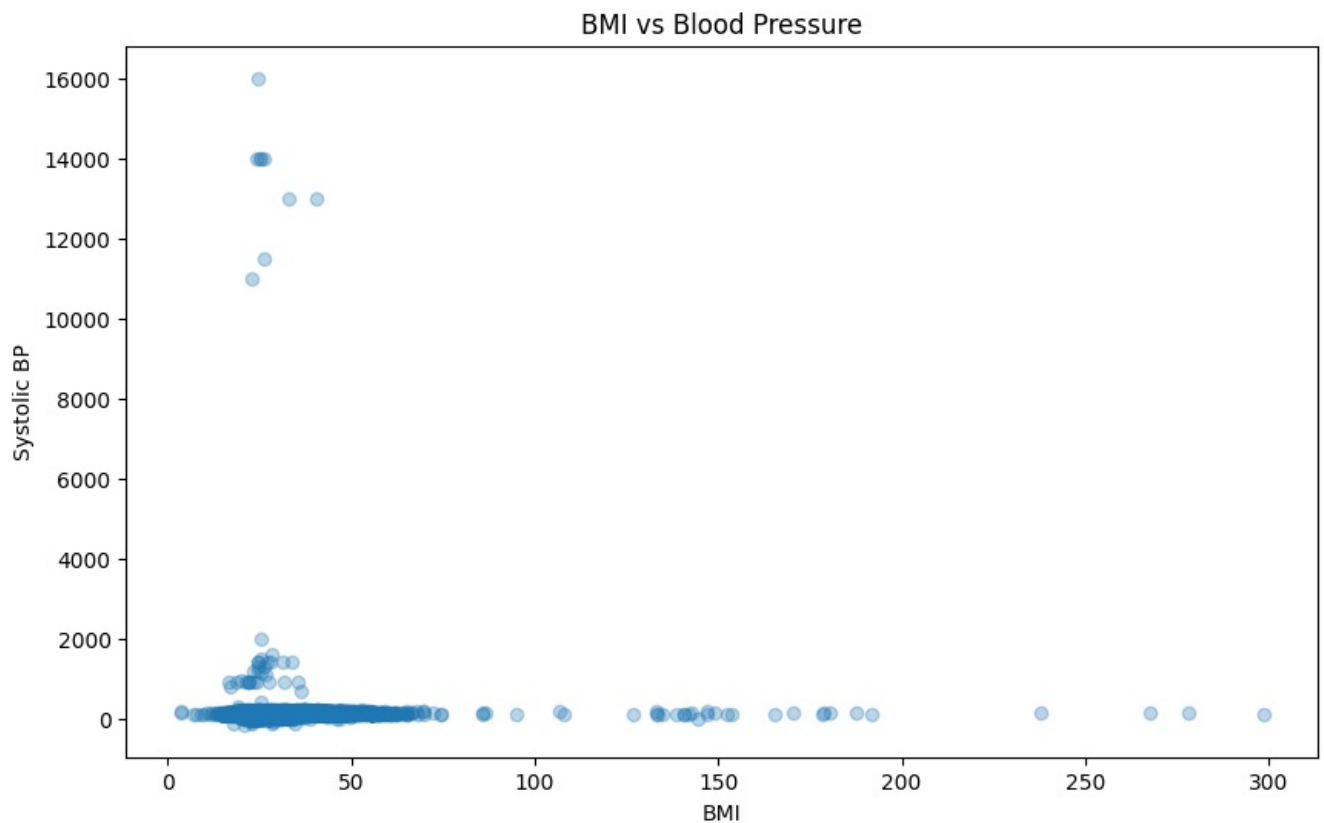
df['BMI_Category'] = df['BMI'].apply(bmi_category)
```

```
In [15]: sns.countplot(x='BMI_Category', data=df)
plt.title("BMI Category Distribution")
plt.show()
```



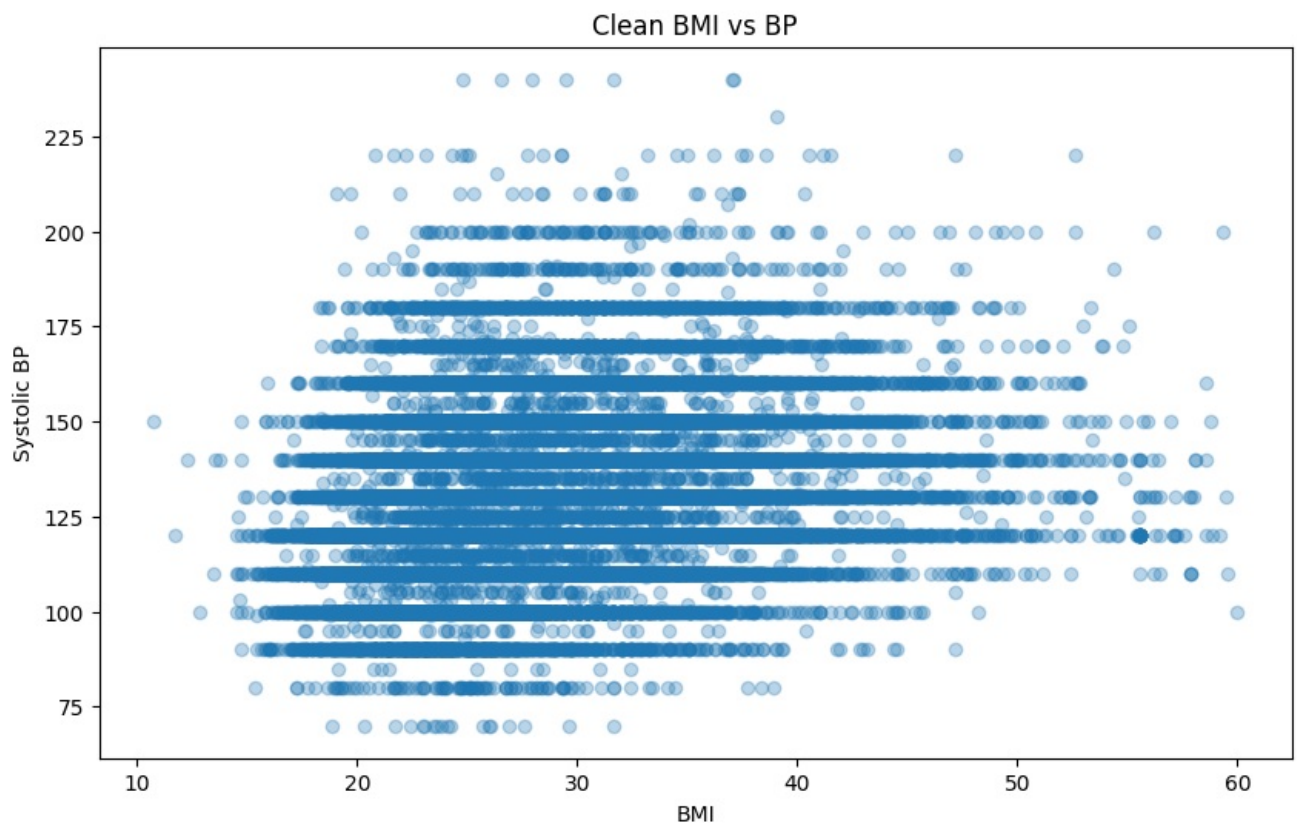
```
In [16]: plt.figure(figsize=(10,6))
```

```
plt.scatter(df['BMI'], df['ap_hi'], alpha=0.3)
plt.xlabel("BMI")
plt.ylabel("Systolic BP")
plt.title("BMI vs Blood Pressure")
plt.show()
```



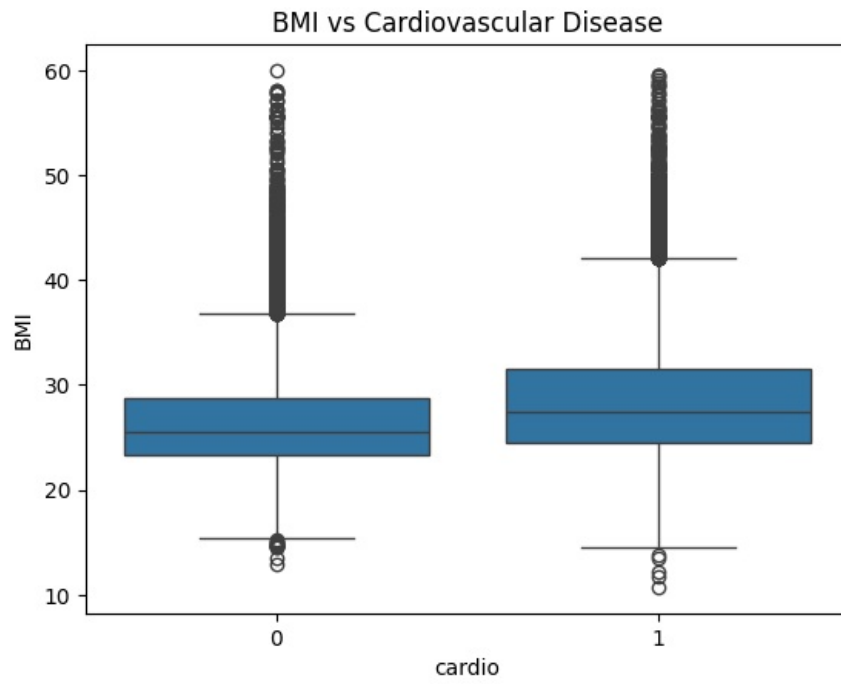
```
In [17]: clean_df = df[(df['BMI'] >= 10) & (df['BMI'] <= 60)]
clean_df = clean_df[(clean_df['ap_hi'] >= 70) & (clean_df['ap_hi'] <= 250)]
```

```
In [18]: plt.figure(figsize=(10,6))
plt.scatter(clean_df['BMI'], clean_df['ap_hi'], alpha=0.3)
plt.xlabel("BMI")
plt.ylabel("Systolic BP")
plt.title("Clean BMI vs BP")
plt.show()
```



```
In [19]: sns.boxplot(x='cardio', y='BMI', data=clean_df)
```

```
plt.title("BMI vs Cardiovascular Disease")  
plt.show()
```



In []: